SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0603 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Case Study

Code of Conduct:

I Mohammed Mudhassir A/192424367 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Mudhassir

1. Intelligent Hospital Disease Diagnosis

Aim: To design a probabilistic system for diagnosing diseases from symptoms and medical history.

Model:

- Use **Bayes' Theorem** to calculate the probability of a disease given observed symptoms:

$$P(D \mid S) = \frac{P(S \mid D)P(D)}{P(S)}$$

- Use **Naïve Bayes** when symptoms are assumed conditionally independent.
- Use a **Bayesian Belief Network (BBN)** when dependencies exist between symptoms.

Example:

Disease → Fever, Cough, Body Pain

The model first learns prior disease probabilities from medical records and then updates them when symptoms are observed.

Analysis: Probability handles uncertainty because symptoms do not always guarantee a disease. BBNs represent relationships between symptoms, while Naïve Bayes provides fast classification.

Real-world constraints: Missing medical data, noisy records, correlated symptoms, and limited training data can affect accuracy.

Conclusion: A probabilistic diagnosis system can provide useful disease probabilities and support doctors in decision-making, but it should be used as a decision-support tool rather than a replacement for medical judgment.

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2. Email Spam Detection

Aim: To classify emails as **Spam** or **Not Spam**.

Model:

- Extract features such as words, links, sender information, and frequency of suspicious terms.
- Train a **Naïve Bayes classifier**.
- Calculate:

$$P(Spam \mid Words)$$

and

$$P(NotSpam \mid Words)$$

- Choose the class with the higher probability.

A **Bayes Optimal Classifier** considers predictions from multiple possible hypotheses and selects the class with the lowest expected error.

MLE vs MDL:

- **Maximum Likelihood Estimation (MLE):** chooses parameters that best fit the observed training data.
- **Minimum Description Length (MDL):** selects a model that provides a good balance between model complexity and data fit.

Analysis: MLE may overfit when the dataset is small. MDL prefers simpler models and can improve generalization.

Conclusion: Naïve Bayes is fast and scalable for spam detection, while MDL can help control model complexity.

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3. Customer Segmentation Using EM

Aim: To identify hidden customer groups from purchasing behavior with incomplete information.

Procedure:

1. Collect customer features such as purchase frequency, amount, and product categories.
2. Assume the data comes from a mixture of probability distributions.
3. Initialize cluster parameters.
4. **E-step:** calculate the probability that each customer belongs to each cluster.
5. **M-step:** update cluster parameters.
6. Repeat until convergence.

Analysis: EM can handle hidden cluster membership and incomplete information. However, different initializations can produce different results because EM may converge to a **local optimum**.

Applications:

- Premium customer identification
- Personalized offers
- Customer retention
- Targeted advertising

Conclusion: EM is useful for discovering hidden customer segments, but multiple initializations and proper validation are required for reliable results.

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4. Medical Image Concept Learning

Aim: To classify medical images as **Normal** or **Abnormal** using concept learning.

Approach:

- Extract features from labeled images.
- Define a hypothesis space containing possible classification rules.
- Use **Candidate Elimination** to maintain:
  - Specific boundary S
  - General boundary G
- Update the boundaries using positive and negative examples.

Finite vs Infinite Hypothesis Space:

- A **finite hypothesis space** contains a limited number of possible rules and is easier to search.
- An **infinite hypothesis space** can represent more complex concepts but requires more data and computation.

Sample complexity: More complex hypothesis spaces require more training examples.

Mistake Bound: It represents the maximum number of mistakes a learner may make before identifying a consistent hypothesis under the learning assumptions.

Conclusion: Candidate Elimination can learn concepts from labeled examples, but generalization depends heavily on the quality and quantity of training data.

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5. Investment Prediction Under Uncertainty

Aim: To predict investment outcomes while continuously updating beliefs.

Approach:

- Define possible outcomes such as **High Return, Moderate Return, and Loss**.
- Assign prior probabilities using historical knowledge.
- Use **Bayesian inference** to update probabilities when new market data arrives.
- Use the **Gibbs algorithm** to sample from complex posterior distributions.

$$P(H \mid D) \propto P(D \mid H)P(H)$$

Analysis: Prior knowledge influences predictions, especially when little data is available. As new data arrives, the posterior probability is updated.

Advantages:

- Represents uncertainty explicitly.
- Supports dynamic decision-making.
- Provides risk probabilities.

Challenges: Gibbs sampling can be computationally expensive and may converge slowly.

Conclusion: Bayesian learning provides a flexible framework for investment decisions, but computational efficiency and reliable prior selection are important.

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6. Autonomous Vehicle Decision System

Aim: To make safe driving decisions under uncertain conditions.

Model:
Use sensor information such as:

- Weather
- Traffic
- Pedestrian movement
- Vehicle speed
- Road conditions

A Bayesian model calculates probabilities for actions such as:

Stop → Slow Down → Continue → Change Lane

Naïve Bayes can classify situations quickly, while Bayesian reasoning updates predictions when new sensor information arrives.

Uncertainty handling: Sensor noise is handled using probability distributions instead of treating sensor readings as perfectly accurate.

Analysis: The system must continuously update its beliefs because traffic and pedestrian behavior change rapidly.

Conclusion: Bayesian probabilistic reasoning can improve decision-making under uncertainty, but autonomous systems require extensive testing because incorrect predictions can have serious consequences.

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7. Chronic Disease Risk Prediction

Aim: To predict patient risk using historical and lifestyle information.

Features:

- Age
- Blood pressure
- Exercise
- Diet
- Medical history
- Family history

Approach:

- Use probability learning to estimate disease probabilities.
- Apply Bayes' Theorem to update risk.
- Use hypothesis space search to identify useful classification rules.
- Consider dependencies between features.

Analysis: Prior probability is important because diseases have different prevalence rates. Larger datasets generally produce more reliable probability estimates.

Interpretability: Bayesian models can provide probability-based explanations, making the results easier to understand.

Conclusion: The model can identify high-risk patients early and support preventive healthcare strategies, but biased or insufficient data can reduce reliability.

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8. Fraud Detection in Banking

Aim: To detect fraudulent transactions while reducing false positives.

Model:
A **Bayesian Belief Network** can represent relationships between:

- Transaction amount
- Location
- Time
- Device
- Customer behavior
- Fraud status

EM can estimate hidden variables such as an unknown customer behavior pattern when some information is missing.

Procedure:

1. Collect transaction data.
2. Build the Bayesian network.
3. Estimate probabilities.
4. Use EM for incomplete data.
5. Calculate the probability of fraud.

Analysis: Probability-based reasoning allows the system to express uncertainty instead of making only a fixed yes/no decision.

Challenges: EM can converge to local optima, and rapidly changing fraud patterns require model updates.

Conclusion: Bayesian methods can improve fraud detection and reduce false alarms by considering multiple dependent factors.

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9. E-Commerce Product Recommendation

Aim: To recommend products based on browsing and purchasing behavior.

Features:

- Previous purchases
- Browsing history
- Product category
- Search terms
- Ratings

Using Naïve Bayes, the system estimates:

$$P(Product \mid UserFeatures)$$

Products with higher probabilities are recommended.

Analysis: Conditional probabilities help determine how user behavior relates to product preferences. Uncertainty is handled by assigning probabilities rather than assuming a user definitely likes a product.

Scalability: Naïve Bayes is computationally efficient and can process large datasets. However, very large product catalogs require efficient feature storage and updating.

Conclusion: Bayesian recommendation can provide fast and scalable personalized recommendations, but user preferences change over time and must be continuously updated.

10. Climate Change Prediction

Aim: To predict environmental changes using historical and incomplete data.

Features:

- Temperature
- Rainfall
- Humidity
- Atmospheric pressure
- CO_2 levels

Approach:

- Use Bayesian inference to represent uncertainty.
- Use prior climate knowledge.
- Use EM to estimate missing values and hidden patterns.
- Repeat EM steps until the likelihood converges.

Analysis: Bayesian models can provide probability distributions rather than a single prediction. EM helps when environmental measurements are incomplete.

Different initializations may cause EM to converge to different local optima.

Conclusion: Probabilistic models are useful for long-term climate forecasting because environmental measurements contain noise and uncertainty. However, long-term predictions remain sensitive to data quality and model assumptions.

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11. Customer Churn Prediction

Aim: To predict whether a customer will leave a service.

Features:

- Monthly usage
- Number of complaints
- Subscription duration
- Service feedback
- Payment history

Approach:

- Use Naïve Bayes to calculate:

$$P(Churn \mid Features)$$

- Use Bayesian inference to update predictions with new customer information.
- Use hypothesis space search to identify useful rules.

Analysis: Prior probability represents the historical churn rate. Dependencies between features can affect predictions because customer behavior variables may be related.

Uncertainty: Instead of simply saying "customer will leave," the model can give a probability such as 0.80 churn risk.

Conclusion: The model can identify high-risk customers and help companies provide targeted offers or support to improve retention.

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12. Smart Healthcare Monitoring

Aim: To detect abnormal vital signs using wearable sensors.

Features:

- Heart rate
- Blood pressure
- Temperature
- Oxygen saturation
- Respiratory rate

BBN Example:

Heart Condition → Heart Rate
Heart Condition → Blood Pressure
Heart Condition → Oxygen Level

The BBN represents dependencies between physiological variables.

Uncertainty: Sensor readings may contain noise or missing values. Bayesian probability allows the system to estimate the likelihood of an abnormal condition despite uncertain readings.

Analysis: Real-time monitoring requires fast inference and continuous updates.

Challenges:

- Sensor noise
- Battery limitations
- Missing data
- False alarms
- Real-time processing

13. Dynamic Route Optimization

Aim: To find efficient delivery routes under uncertain traffic and weather.

Approach:

- Collect historical traffic and weather data.
- Calculate prior probabilities for traffic conditions.
- Use Bayesian inference to update these probabilities using real-time information.
- Calculate the expected travel time and choose the route with the best expected outcome.

For example:

$$P(Traffic \mid CurrentData)$$

can be used to estimate the probability of congestion.

Analysis: Historical data provides prior knowledge, while GPS and traffic updates modify the prediction dynamically.

Challenges:

- Large search space
- Real-time computation
- Changing traffic
- Uncertain weather

Conclusion: Bayesian route optimization can adapt to changing conditions, but efficient algorithms are required for real-time decision-making.

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14. Cybersecurity Intrusion Detection

Aim: To detect normal and abnormal network activity.

Features:

- Number of connections
- Packet size
- Protocol
- Connection duration
- Failed login attempts

Approach:

- Use Naïve Bayes to classify traffic as normal or intrusion.

- Use EM to identify hidden patterns in unlabeled network traffic.
- Use Bayesian learning to update probabilities when new attacks are detected.

Analysis: EM can handle incomplete information and discover hidden attack patterns. Bayesian learning allows the system to adapt to changing network behavior.

False alarms: Probability thresholds can be adjusted to balance intrusion detection and false positives.

Conclusion: Combining Naïve Bayes, EM, and Bayesian learning provides an adaptive intrusion detection system suitable for dynamic network environments.

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15. Crop Yield Prediction

Aim: To predict crop yield under uncertain environmental conditions.

Features:

- Rainfall
- Temperature
- Soil quality
- Fertilizer usage
- Irrigation
- Historical crop yield

Approach:

- Use historical agricultural data to establish prior probabilities.
- Apply Bayesian inference to calculate yield predictions.
- Update the prediction when new seasonal information becomes available.

$$P(Y\text{'}ield \mid WeatherSoil)$$

Analysis: Prior agricultural knowledge improves predictions when current data is limited. Seasonal variations can significantly change rainfall, temperature, and crop yield probabilities.

Reliability: The model should be evaluated using historical test data and uncertainty intervals rather than relying only on a single predicted yield.

Applications:

- Crop planning
- Irrigation decisions
- Fertilizer management